Win Back Model
(Health Insurance)

Approach Note

24 Mar, 2025

Objective

The winback model is to predict the likelihood of previously churned health insurance customers re-engaging with the company. By leveraging the XGBoost algorithm, the model aims to identify high-potential lapsed customers based on historical behavioural, demographic, and policy-related data. This enables marketing and retention teams to prioritize outreach efforts, personalize communication, and allocate resources more efficiently—ultimately increasing the success rate of customer winback campaigns and reducing customer acquisition costs.

Customer base considered for modeling:

- CSV (Customer Single View) dataset (1st April 2019 to 27th August 2022)

Approach:

- Our **Winback Model** is designed to assign probabilities to previously churned customers based on their likelihood of re-engaging with our health insurance services. A customer is considered **churned** if they fail to renew their policy within a **3-month grace period** following the policy's expiry. These are users who were once active policyholders but did not renew within this timeframe.
- Using historical data, including past user behavior, demographic details, policy features, claim history, and prior interaction patterns, the model learns distinguishing patterns between customers who returned (won back) and those who did not.
- Based on the field available in the dataset, we create a column named **Target**, where a value of 1 denotes a **won-back user** and 0 denotes a **non-returning user**. The model is trained to predict this binary outcome and assigns probabilities to both classes (0,1).
- Those predicted as class 1 (i.e., high winback likelihood) with higher probabilities are considered prime targets for winback campaigns. These users are more likely to respond positively to re-engagement efforts and are prioritized accordingly.

Outliers and exclusions:

- Columns where missing values were more than 50% were excluded from the modeling process.
- Missing values for numerical columns were filled with median values and for categorical values were filled with mode values.
- Binning is done by using bivariate analysis to assess the relationship between continuous variables and the target. This helps group variables into discrete bins, reducing outlier impact and enhancing model performance.

Modeling:

- We used XGBoost for the Winback Model to predict the likelihood of churned customers re-engaging with the health insurance service.
- To address the class imbalance in the dataset, we calculated the scale_pos_weight by determining the ratio of the majority class (non-returning users) to the minority class (won-back users). This was done by counting the number of instances in each class and dividing the majority class count by the minority class count. This scale_pos_weight value was then incorporated into the XGBoost model, enabling it to give more importance to the minority class (won-back users), improving the model's ability to predict re-engaged customers accurately.
- After training the model on the entire dataset, we evaluated feature importance using XGBoost's output. Any features with an importance score below 1.00 were excluded, ensuring that only the most significant features contributed to the final model.
- Variables for modelling

Fields taken	Usage in modeling
POLICYNO	Not used due to being identifier
GENDER	Not used due to being identifier
OCCUPATION_ORIGINAL	Used
MARITALSTATUS	Feature importance less than 1%
ISMOBLIE	Not used due to insignificance
ISEMAIL	Not used due to insignificance
STATE	Used
PINCODE	Not used due to insignificance
POLICYSTARTDATE	Not used due to insignificance
POLICYENDDATE	Not used due to insignificance
POLICYVARIANT	Feature importance less than 1%
PRODUCT NAME ORIGINAL	Used
PRODUCT CODE ORIGINAL	Not used due to insignificance
POLICYTENURE	Feature importance less than 1%
PLAN TYPE ORIGINAL	Used
STP / NSTP	Used
ISCHRONIC	Not used due to insignificance
NET_PREMIUM	Feature importance less than 1%
SUM_INSURED	Not used due to insignificance
SMOKINGCOUNT	Not used due to large percentage of missing values
PLAN ORIGINAL	Not used due to large percentage of missing values
AGE	Feature importance less than 1%
CLAIM_COUNT	Not used due to large percentage of missing values
HRCNT	Not used due to large percentage of missing values
ACTIVEDAYS ORIGINAL	Not used due to insignificance
ACTIVEDAYS_CNT	Feature importance less than 1%
HHS_CNT	Used
HA_CNT	Not used due to insignificance
APPLOGIN_CNT	Feature importance less than 1%
ADULT_CNT	Used
CHANNEL	Used

HHS	Not used due to large percentage of missing values
BUSINESSTYPE	Used
PORTABILITYFLAG	Used
NUMBER_OF_MEMBERS	Feature importance less than 1%
BMI	Used
REPORTING_YEAR	Used
POLICY_VINTAGE_REVISED	Feature importance less than 1%
TARGET	Target Variable
DATEDIFFRENEWAL	Not used due to large percentage of missing values
DATEDIFF_MONTH	Not used due to large percentage of missing values
RENEWAL_Y/N	Not used due to large percentage of missing values

Performance:

Classification Report (Validation):

	precision	recall	f1-score	support
0	0.98	0.6	0.74	190975
1	0.12	0.8	0.21	13251
accuracy			0.61	204226
macro avg	0.55	0.7	0.48	204226
weighted avg	0.92	0.61	0.71	204226

Challenges:

- The dataset is heavily skewed toward class 0, which makes it harder for the model to learn meaningful patterns for class 1.
- This imbalance leads to high accuracy overall, but low performance on the minority class.
- The model may be biased toward class 0 due to its overwhelming presence in the training data, which may lead to overfitting to Majority Class

Conclusion:

- The model shows strong precision in identifying the majority class, which makes up most of the data, and achieves an overall accuracy of 61%.
- For the minority class, the model captures most relevant cases well (recall of 80%), which is a good sign. However, precision is lower, meaning there's a higher rate of false positives.